Electrode Atomic DEVS Model in Cadmium v2

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SYSC 5104 – Assignment 1

Photo credit: ChatGPT

1 Electrode–Brain DEVS Model

I model a simplified neural environment around a recording electrode in the brain. The brain produces electric activity that the electrode can measure. If the combined electrical activity (voltage) crosses some predefined threshold, the electrode sends a digital signal, suggesting some nearby neuron has likely fired (a.k.a. "spiked"). Otherwise, the electrode stays passive.

The system is decomposed into the following components, as shown in Figure 1:

- **Electrode (atomic DEVS model)**: observes the summed local potential and outputs a 1 if the potential is above a threshold (default 25 mV) and is passive otherwise.
- **Neurons (identical atomic DEVS models)**: idealized spiking neurons with a resting potential of $V_{\text{rest}} = -70\,\text{mV}$ and a spike potential of $V_{\text{spike}} = 30\,\text{mV}$. Each neuron emits spikes at random times and is otherwise clamped at rest. After emitting a spike, a neuron enters a refractory state of duration $T_{\text{ref}} = 2.0\,\text{s}$ during which it cannot fire again. In this simulation, the system is composed of two neurons.
- **Background noise (atomic DEVS model)**: a stochastic process representing ongoing synaptic and other activity in the tissue. It produces Gaussian noise with mean $\mu = 0\,\text{mV}$

and standard deviation $\sigma = 10\,\mathrm{mV}$ at a fixed temporal resolution (set to $\Delta t = 0.1\,\mathrm{s}$ in this simulation).

- **Adder (atomic DEVS model)**: sums background noise and neuron contributions. The neuron contributions are attenuated in accordance with the distance of the neurons from the electrode. In this simuation, the attenuation factors are set to 0.9 and 0.75 for Neuron 1 and Neuron 2, respectively.
- **Brain (coupled DEVS model)**: The brain is composed of the background noise model, the models of neurons, and the adder model.
- **Top (coupled DEVS model)**: the top model is composed of the electrode and the brain models.

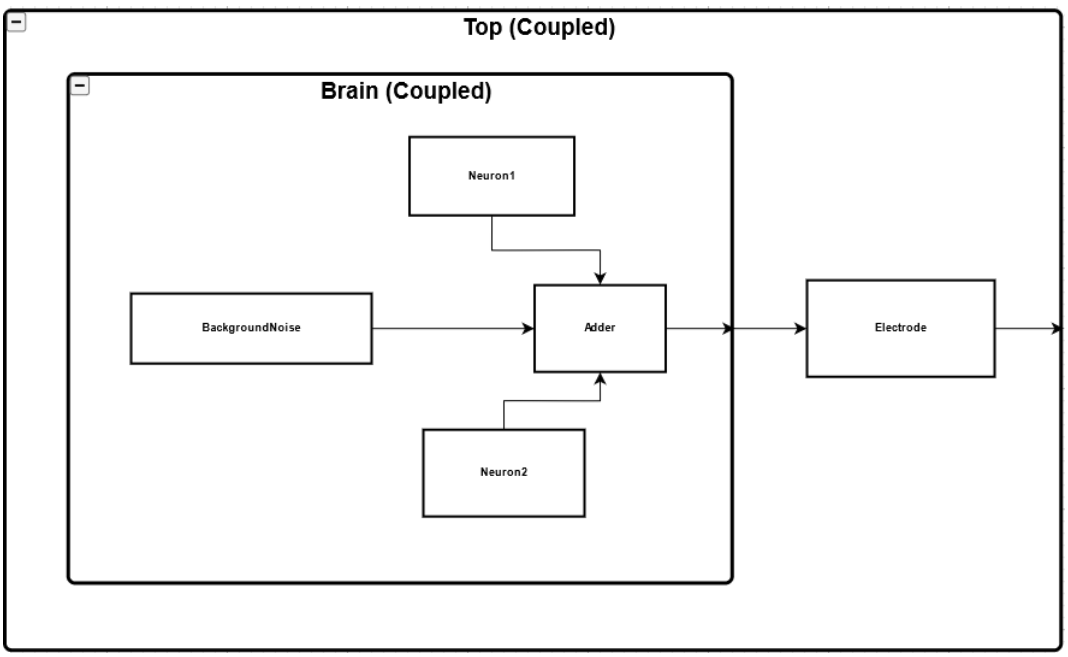


Figure 1: DEVS-Graph representation of the electrode–brain model, showing the coupled hierarchy and port-to-port couplings (generated with DEVS-Graph).

2 Coupled Model Specifications

2.1 Top coupled model.

The top model is

$$M_{\text{top}} = \langle X_{\text{top}}, Y_{\text{top}}, D_{\text{top}}, \{M_d\}_{d \in D_{\text{top}}}, EIC, EOC, IC \rangle,$$

with

$$\begin{aligned} X_{\text{top}} &= \emptyset, \\ Y_{\text{top}} &= \{y_{\text{spike}}\}, \quad y_{\text{spike}} \in \{1\}. \end{aligned}$$

The component set is

$$D_{\text{top}} = \{\text{brain}, \text{electrode}\},$$

where M_{brain} is the brain coupled model and $M_{\text{electrode}}$ is the electrode atomic model. There are no external input couplings (EIC= $\emptyset$). The internal coupling and external output coupling are:

$$\begin{aligned} IC &= \{(\text{brain}, \texttt{brain_out}) \to (\text{electrode}, \texttt{input_in})\}, \\ EOC &= \{(\text{electrode}, \texttt{spike_out}) \to (M_{\text{top}}, y_{\text{spike}})\}. \end{aligned}$$

2.2 Brain coupled model.

The brain model is

$$M_{\text{brain}} = \langle X_{\text{brain}}, Y_{\text{brain}}, D_{\text{brain}}, \{M_d\}_{d \in D_{\text{brain}}}, EIC, EOC, IC \rangle,$$

with

$$\begin{aligned} X_{\text{brain}} &= \emptyset, \\ Y_{\text{brain}} &= \{y_{\text{brain}}\}, \quad y_{\text{brain}} \in \mathbb{R}. \end{aligned}$$

The component set is

$$D_{\text{brain}} = \{\text{noise}, \text{neuron1}, \text{neuron2}, \text{adder}\}.$$

Couplings:

$$\begin{aligned} IC = \{&(\text{noise}, \texttt{noise_out}) \to (\text{adder}, \texttt{noise_in}), \\ &(\text{neuron1}, \texttt{potential_out}) \to (\text{adder}, \texttt{neuron1_in}), \\ &(\text{neuron2}, \texttt{potential_out}) \to (\text{adder}, \texttt{neuron2_in})\}, \\ EOC = \{&(\text{adder}, \texttt{sum_out}) \to (M_{\text{brain}}, y_{\text{brain}})\}, \\ EIC = \emptyset&. \end{aligned}$$

3 Atomic Model Specifications

Each atomic model is specified as a classical DEVS

$$M = \langle X, Y, S, \delta_{\text{int}}, \delta_{\text{ext}}, \delta_{\text{con}}, \lambda, ta \rangle.$$

3.0.1 Background noise model

Ports and state.

$$\begin{aligned} X_{\text{noise}} &= \emptyset, \\ Y_{\text{noise}} &= \{y_n\}, \quad y_n \in \mathbb{R}, \\ S_{\text{noise}} &= \{(v, \sigma) \mid v \in \mathbb{R},\ \sigma \in \mathbb{R}_+\}, \end{aligned}$$

where v is the current noise sample and σ is the time remaining to the next sample.

Time advance.

$$ta(s) = \sigma, \quad \text{with initial } \sigma_0 = 0.1.$$

Internal transition. At each internal event:

$$v' \sim \mathcal{N}(0,\, 10^2),$$
$$\sigma' = 0.1.$$

External and confluent transition. There are no inputs, so δ_{ext} is trivial; δ_{con} reduces to applying δ_{int} followed by an immediate adjustment of elapsed time.

Output.

$$\lambda(s) = v.$$

3.0.2 Neuron model

The neuron model represents a stochastic spike generator. Unlike the original formulation, the neuron does not continuously output its membrane potential. Instead, it emits an output event only when a spike occurs.

Ports and state.

$$\begin{aligned} X_{\text{neuron}} &= \emptyset, \\ Y_{\text{neuron}} &= \{y_\nu\}, \quad y_\nu \in \mathbb{R}, \\ S_{\text{neuron}} &= \{(\sigma, r, f) \mid \sigma \in \mathbb{R}_+,\ r \in \mathbb{R}_+,\ f \in \{0,1\}\}. \end{aligned}$$

Here

- σ is the time until the next evaluation event,
- r is the remaining refractory time,
- f indicates whether the neuron fires at the current event.

Time advance

$$ta(s) = \sigma, \qquad \sigma_0 = 0.1.$$

Thus the neuron evaluates its firing condition every 0.1 s.

Internal transition Let $\Delta t = \sigma$. If the neuron is currently in its refractory period, the refractory timer is decreased:

$$r' = \max(0, r - \Delta t).$$

If $r' > 0$, the neuron cannot fire and the firing flag is set to zero.
Otherwise a random variable

$$U \sim \mathcal{U}(0,1)$$

is drawn. If $U < p$, where p is the firing probability, the neuron fires and

$$f' = 1, \qquad r' = T_{\text{ref}}.$$

Otherwise the neuron remains silent:

$$f' = 0.$$

In all cases the next evaluation time is scheduled with

$$\sigma' = 0.1.$$

Output The neuron produces an output event only when a spike occurs:

$$\lambda(s) = \begin{cases} V_{\text{spike}} & \text{if } f = 1, \\ \text{no output event} & \text{otherwise.} \end{cases}$$

External and confluent transitions Since the neuron has no inputs, δ_{ext} is trivial and δ_{con} reduces to the internal transition.

3.0.3 Adder model

Ports and state.

$$\begin{aligned} X_{\text{adder}} &= \{x_n, x_1, x_2\}, \quad x_i \in \mathbb{R}, \\ Y_{\text{adder}} &= \{y_a\}, \quad y_a \in \mathbb{R}, \\ S_{\text{adder}} &= \{(n, v_1, v_2, s, \sigma)\}, \end{aligned}$$

where n is the latest noise value, v_1 and v_2 are the neuron potentials, s is the current sum, and σ is time to next internal event.

Time advance. The adder is passive until it receives inputs:

$$ta(s) = \sigma,$$

with initial $\sigma = +\infty$.

External transition. When any input arrives, the adder updates the corresponding value. Background noise is updated whenever a noise message arrives. Neuron inputs are interpreted as instantaneous spike events. If no neuron message arrives at the current time step, the neuron contribution is taken to be zero.

The summed potential is computed as

$$s = n + 0.9\, v_1 + 0.75\, v_2.$$

An immediate internal event is scheduled by setting $\sigma' = 0$.

Internal transition. After emitting an output at t, the adder returns to the passive state:

$$\sigma' = +\infty$$

with the current n, v_1, v_2, s preserved.

Output.

$$\lambda(s) = s.$$

3.0.4 Electrode model

Ports and state.

$$
\begin{aligned}
X_{\text{elec}} &= \{x_e\}, \quad x_e \in \mathbb{R}\ (\text{mV}),\\
Y_{\text{elec}} &= \{y_e\}, \quad y_e \in \{1\},\\
S_{\text{elec}} &= \{(u,\theta,\sigma) \mid u \in \mathbb{R},\ \theta \in \mathbb{R},\ \sigma \in \mathbb{R}_{\geq 0} \cup \{\infty\}\}.
\end{aligned}
$$

Here u is the most recent input potential, θ is the detection threshold, and σ is the time until the next internal event.

Time advance. The electrode is passive by default:

$$ta(s) = \sigma, \qquad \sigma_0 = \infty.$$

External transition. Upon receiving an input x_e after elapsed time e:

$$\delta_{\text{ext}}((u,\theta,\sigma), e, x_e) = \begin{cases} (u \leftarrow x_e,\ \theta,\ 0), & x_e \geq \theta,\\ (u \leftarrow x_e,\ \theta,\ \infty), & x_e < \theta. \end{cases}$$

That is, the electrode schedules an immediate internal event only when the threshold condition is satisfied.

Internal transition. After producing a spike, the electrode returns to the passive state:

$$\delta_{\text{int}}(u,\theta,0) = (u,\theta,\infty).$$

Output. Output occurs only when $\sigma = 0$ (i.e., only when $u \geq \theta$):

$$\lambda(u,\theta,0) = 1.$$

Confluent transition. When an internal and external event coincide, the implementation prioritizes processing the external input and schedules an immediate output only if the threshold is met.

4 Experimentation Framework

In this section, I propose and implement an experimentation framework for the electrode–brain DEVS model. The framework follows the model hierarchy and expands tests incrementally. All experiments are implemented in Cadmium v2.

4.1 Atomic Model Experiments

4.1.1 Background noise

The `BackgroundNoise` atomic model is tested with the driver `test_background_noise.cpp`, which implements the following experiment:

Experiment N1: Sampling interval and distribution test. This experiment simulated the `BackgroundNoise` model for $T = 200.0\,\text{s}$, and its outputs were logged to `noise_sampling_log.csv`. The expected behavior is that internal events (and thus outputs) occur regularly at $t = 0.0, 0.1, \ldots, 200.0\,\text{s}$, corresponding to the specified time advance of $ta(s) = 0.1\,\text{s}$. Thus, there should be 2000 outputs generated during this experiment. Moreover, the outputs ought to follow a normal distribution with $\mathbb{E}[v] \approx 0\,\text{mV}$ and $\text{std}(v) \approx 10\,\text{mV}$, validating the Gaussian noise assumption used in the model.

The resulting statistical properties of the generated noise are:

Number of samples	2000
Mean	0.122 mV
Standard deviation	9.94 mV

These values are consistent with the intended Gaussian noise model $\mathcal{N}(0, 10^2)$.

To further verify normality, the Shapiro–Wilk test was applied:

Test statistic	0.9987
p-value	0.141

Since the p-value is greater than 0.05, we cannot reject the hypothesis that the data were generated from a normal distribution (in other words, the data follows the normal distribution, as expected).

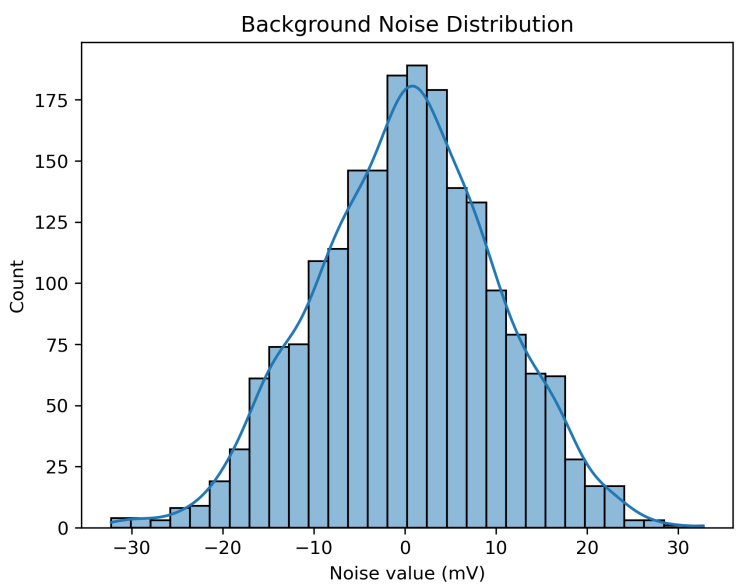
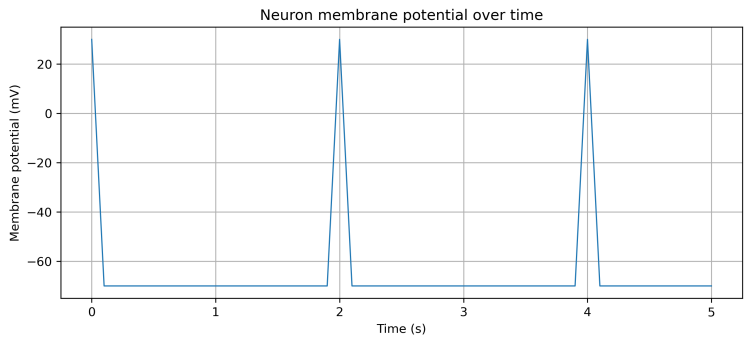


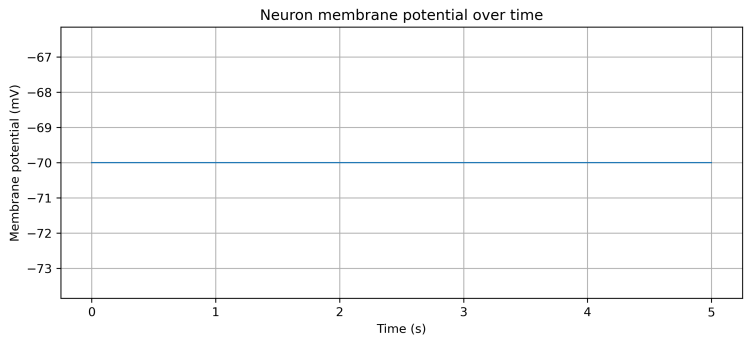
Figure 2: Normal Q–Q plot of the background noise samples.

Experiment Ne1: Always firing neuron. The neuron is initialized with firing probability $p = 1.0$ and simulated for $T = 5\,\text{s}$. The neuron produces spike events of $V_{\text{spike}} = 30\,\text{mV}$ whenever it is not in the refractory state. After each spike, the neuron enters the refractory period $T_{\text{ref}} = 2.0\,\text{s}$ during which no output events occur and the neuron remains at $-70\,\text{mV}$.

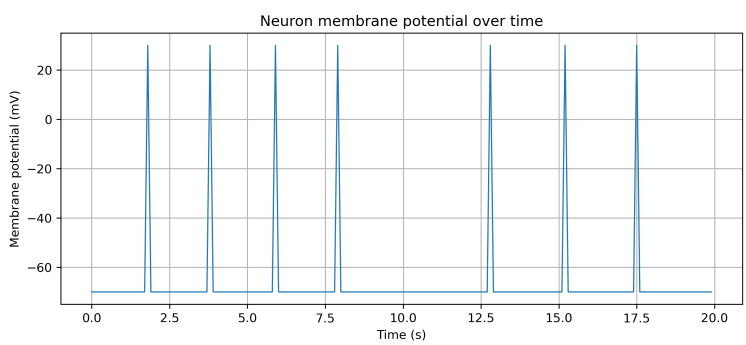
The model performed as expected, generating pulses every 2.0 s:



Experiment Ne2: Never firing neuron. With $p = 0.0$, the neuron never produces spike events. The output log therefore contains no neuron messages, validating that the model remains silent when the firing probability is zero:



Experiment Ne3: Default firing probability. With the default firing probability $p = 0.1$, the neuron is simulated for $T = 20\,\text{s}$. The resulting trace contains sparse spike events separated by refractory intervals, representing stochastic neuronal firing:



4.1.2 Adder

The `Adder` atomic model is tested using `test_adder.cpp`. In this experiment, the two `Neuron` models and the `Background Noise` model are connected to the `Adder`.

Experiment A1: Addition verification. The experiment logs the input received by the `Adder` and logs the output, checking that the inputs are summed correctly. The resulting sum is logged

in `adder_simple_log.csv`. The expected value is

$$s = n + 0.9v_1 + 0.75v_2.$$

Here are a few rows of the data generated, showing that the adder sums inputs as expected:

Table 1: Verification of the Adder model output. The expected sum is $s = n + 0.9v_1 + 0.75v_2$. Missing neuron inputs indicate that no spike event was emitted at that time step.

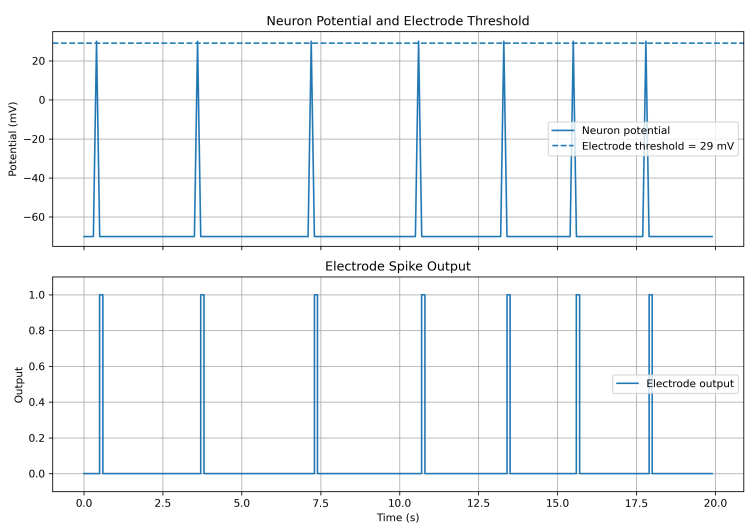
Time (s)	Noise (mV)	v_1 (mV)	v_2 (mV)	Expected sum (mV)	Output (mV)
0.1	0.000000	–	30.0	22.500000	22.500000
0.2	12.809000	30.0	–	39.809000	39.809000
0.3	-4.957100	–	–	-4.957100	-4.957100
0.4	-2.612810	–	–	-2.612810	-2.612810
0.5	3.577180	–	–	3.577180	3.577180

4.1.3 Electrode

The `Electrode` atomic model is event-based: it emits a spike value 1 only when the input potential satisfies $u \geq \theta$. If $u < \theta$, the model remains passive and produces no output event.

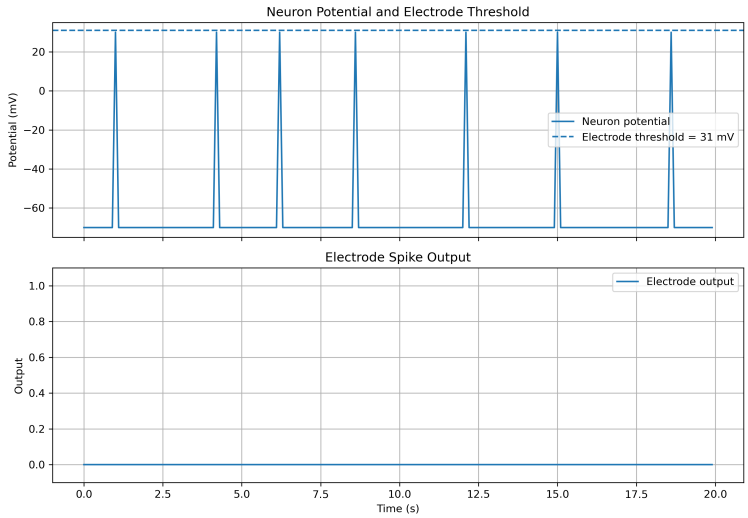
Experiment E1: Threshold crossing In this experiment, the `Electrode` is connected to a single `Neuron` model. With the threshold set below 30 mV (set to 29 mV in the actual experiment), the electrode ought to send a signal any time the neuron fires.

The electrode fires, as expected:



Experiment E2: High threshold This experiment is analogous to Experiment E1, but the threshold is set above 30 mV (set to 31 mV in the actual experiment). The expected behavior is that the electrode should never send any signals.

The electrode never fires, as expected:



4.2 Top-Level Integration Experiments

All models are combined and the complete electrode–brain system is simulated. This is done to verify that all the models have been linked corretly. Moreover, a final verification can be done that all the outputs from neurons and the background noise are correctly added. Toward this end, the following experiments were carried out.

Experiment T1: Threshold $45\,\mathrm{mV}$**.** Both neurons are configured to fire with probability $p = 1.0$, ensuring that spike events occur whenever the neurons are not in their refractory period. Also, the neurons are timed to spike at the same time. The electrode threshold is set to

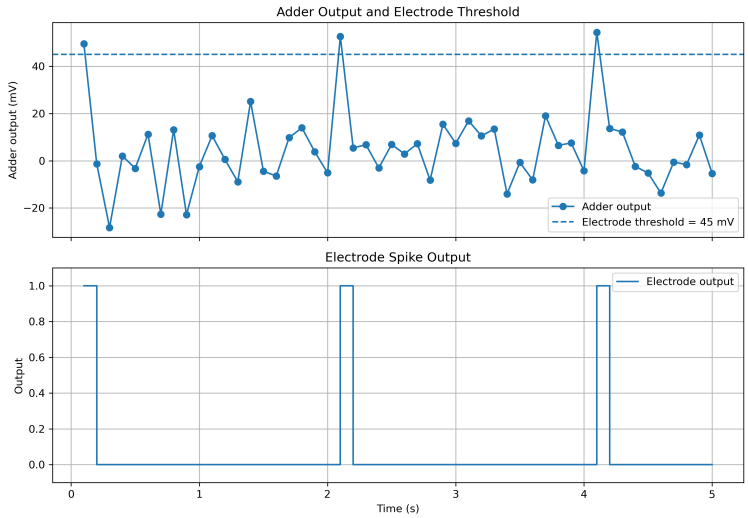
$$\theta = 45\,\mathrm{mV}.$$

Since the attenuated spike contributions are

$$0.9 \cdot 30 + 0.75 \cdot 30 = 49.5\,\mathrm{mV},$$

the summed signal exceeds the electrode threshold. This leads to the eletrode outputing a signal, except in cases when the background noise is less than $-4.5\,\mathrm{mV}$. Results are logged to `top_threshold_45_log.csv`.

The results are as expected, with the electrode firing whenever the combined input signal crosses the threshold set on the electrode:

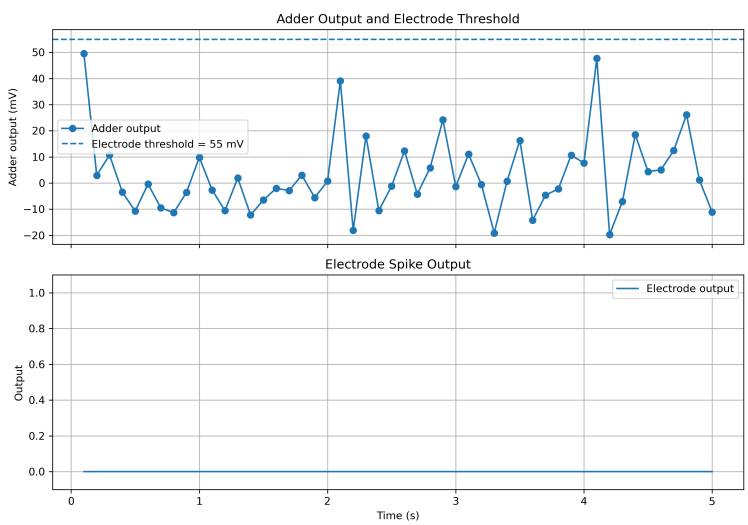


Experiment T2: Threshold 55 mV**.** The same neuron configuration is used, but the electrode threshold is increased to

$$\theta = 55\,\mathrm{mV}.$$

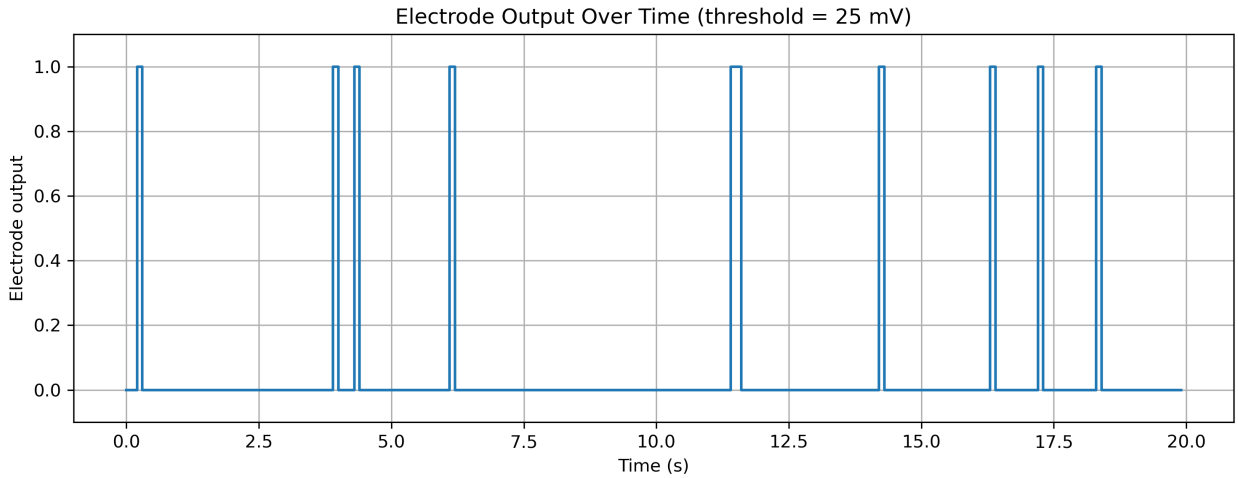
This way, the electrode almost always stays passive, unless noise exceeds 5.5 mV and the neurons spike. Results are logged to `top_threshold_55_log.csv`.

The results are as expected. With the electrode's threshold set high, the combined signal never crosses the threshold and so the electrode never fires:



4.3 Electrode-brain system simulation

With all the hierarchically expanded experimental tests completed, the final simulation of the electrode inside the brain is run with default values of the electrode-brain model. The simulation is ran for $T = 20.0s$, producing the following results:


Electrode Output Over Time (threshold = 25 mV)
Electrode output
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)